

**FLOODSENSE: AN AUGMENTED REALITY MOBILE APPLICATION WITH
MACHINE LEARNING-BASED FLOOD SUSCEPTIBILITY MAPPING FOR
BACOR CITY**

Undergraduate Thesis
Submitted to the Faculty of the
Department of Computer Studies
Cavite State University – Bacoor City Campus
City of Bacoor, Cavite

In partial fulfillment of the
requirements for the degree
Bachelor of Science in Computer Science

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May 2027

FloodSense: An Augmented Reality Mobile Application with Machine Learning-Based Flood Susceptibility Mapping for Bacoar City

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An undergraduate thesis manuscript submitted to the faculty of the Department of Computer Studies, Cavite State University – Bacoar City Campus, City of Bacoar, Cavite, in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science with Contribution No. _____. Prepared under the supervision of

INTRODUCTION

Flooding is one of the most frequent and destructive natural hazards in the Philippines. The country's location within the typhoon belt exposes it to intense rainfall events that cause widespread flooding, displacement, and loss of life (PSA, 2025). Beyond physical destruction, communities that lack accessible, location-specific flood susceptibility information are often unable to make informed decisions before and during flood events, compounding the human cost. Bacoar City, Cavite, is among the localities most acutely affected, situated in a low-lying area of the province, the city faces persistent flood susceptibility driven by heavy seasonal rainfall, proximity to major river systems, and inadequate drainage infrastructure (JICA, 2021).

To assess this challenge, the researchers surveyed 50 residents across 12 barangays in Bacoar City, with respondents drawn largely from historically flood-affected barangays in the city. The findings revealed that while flood warnings are disseminated through SMS, social media, and barangay PA systems, these channels do not provide location-specific susceptibility information before a flood develops. Only 20% of respondents felt well-informed about which areas in the city

are flood-prone meaning 80% fell under Somewhat Aware, Not Sure, or Not Aware at All and most rated the timeliness, clarity, and quality of current flood information as near-dissatisfied. Most critically, 64% reported making a wrong decision taking flooded roads, failing to evacuate, or misjudging flood depth due to the absence of accurate, area-specific information. A Key Informant Interview with the Bacoar City Disaster Risk Reduction and Management Office (BDRRMO) confirmed that while the office maintains historical flood records and GIS-based exposure maps, these datasets remain inaccessible to the public in any interactive format. These findings point to a clear need for a digital tool that bridges the gap between institutional flood data and the communities who need it most.

To bridge this gap, the researchers developed a mobile application that provides residents with accessible, visual flood susceptibility information. According to Marques et al. (2024), mobile augmented reality applications built on Unity and Vuforia can effectively deliver flood impact visualizations through everyday smart devices using color-coded indicators, making it a practical medium for community engagement with spatial hazard information. By integrating AR, users can point their phone camera at any flat surface and view a color-coded susceptibility map of Bacoar City's barangays rendered as a ground-plane overlay transforming institutional flood data into an intuitive, accessible visualization.

The application uses a machine learning model trained on elevation, rainfall parameters, and historical flood records from BDRRMO and CPDCO to classify each barangay into one of four susceptibility levels Low, Moderate, High, and Very High. Rainfall input is simplified through a PAGASA-aligned dropdown with plain-language intensity presets and a duration selection, eliminating the need for technical meteorological knowledge. By coupling this classification model with markerless augmented reality, the application delivers location-specific, color-coded susceptibility maps directly to residents' devices, bridging the gap between institutional data and community awareness. In doing so, the study aims

to provide Bacoor City residents with an intuitive decision-support tool that reduces the wrong decisions, delayed evacuations, and uninformed travel choices that currently occur every flood season.

Statement of the Problem

This study aimed to develop a mobile application that provides barangay-level flood susceptibility information for Bacoor City through machine learning classification and augmented reality visualization. The research sought to answer key questions based on survey data from 50 residents across 12 barangays in Bacoor City, many of which have historically experienced flooding, a key informant interview with the Bacoor Disaster Risk Reduction and Management Office (BDRRMO) and the City Planning and Development Coordinator Office (CPDCO), and a review of related literature.

In Bacoor City, flooding is a persistent and consequential reality. Survey results reveal that 92% of respondents have personally experienced flooding, with 58% describing the flooding as severe (waist-deep or higher). Despite this exposure, only 20% of residents consider themselves well-informed about flood-prone areas, and mean satisfaction scores for information quality, timeliness, and clarity range from 2.62 to 2.72 on a 5-point scale — clustering at the lower end of the Neutral range and trending toward dissatisfaction. These information shortcomings carry direct consequences: 64% of respondents have made wrong decisions due to insufficient flood information, including taking flooded routes, failing to evacuate in time, and misjudging conditions. Open-ended responses further identified the top problems as late or delayed information (26%) and insufficient or absent location-specific warnings (20%). The gap is most acute for residents who face urgent safety decisions with little to no reliable flood information during rainfall events. *How can a mobile application that combines*

machine-learning-based susceptibility classification with augmented reality visualization empower residents to make informed decisions and reduce preventable wrong choices during flood events?

The problem extends to the local disaster risk reduction infrastructure. The BDRRMO maintains historical flood records, GIS-based exposure maps, and barangay-level hazard data; however, these datasets are not accessible to the public in an interactive or on-demand format. Current dissemination relies on social media, email, and barangay coordination methods that the BDRRMO key informant acknowledged do not fully address the public's need for personalized, location-specific guidance. The office further noted that resident engagement with official warnings remains low due to work constraints, poverty, and limited attendance at flood orientations. *How can a system that transforms the BDRRMO's institutional datasets and environmental indicators into an intuitive, user-facing mobile interface serve as a complementary decision-support tool, extending barangay-level flood susceptibility information directly to residents?*

Taken together, these gaps reveal a disconnect between the institutional flood data maintained by the local government and the real-world information needs of Bacoar City's residents. The overarching question this study seeks to answer is: *How can a mobile application integrating PAGASA-aligned rainfall input, machine-learning-based susceptibility mapping, and augmented reality visualization effectively close the gap between institutional flood data and community-level awareness, reduce the incidence of wrong decisions caused by information lack, and provide a locally grounded example of technology-supported disaster risk communication for Philippine cities?*

Objectives of the Study

Generally, the study aimed to develop a mobile application that delivers barangay-level flood susceptibility mapping for Bacoor City through the integration of augmented reality visualization and machine learning classification, thereby enhancing flood awareness and decision-making for residents, and the local disaster risk reduction office.

Specifically, it aimed to:

1. identify the prevailing challenges, pain points, and needs of Bacoor City residents and relevant local government offices concerning flood information access, timeliness, clarity, and decision-making through a structured survey of 50 respondents across 12 barangays, a key informant interview with the Bacoor Disaster Risk Reduction and Management Office (BDRRMO) to validate survey findings and assess institutional data availability, and a supplementary key informant interview with the City Planning and Development Coordinator Office (CPDCO) to identify available geospatial and land use datasets relevant to flood susceptibility modeling all conducted during the Empathize phase of the Design Thinking methodology;
2. analyze existing flood-related datasets — including BDRRMO historical flood records, CPDCO elevation data and land use maps, and PAGASA rainfall classifications — and evaluate supervised machine learning approaches to determine the most effective environmental features and classification model for predicting barangay-level flood susceptibility into four levels (Low, Moderate, High, Very High);
3. develop a mobile application using Unity and Vuforia SDK for markerless augmented reality ground-plane visualization, Firebase for

user authentication and cloud data storage, and Python with scikit-learn/XGBoost for the machine learning classification model, integrating a PAGASA-aligned rainfall input interface, surface detection indicator, user account system, and onboarding disclaimer;

4. test the system's core modules by implementing unit testing, integration testing, machine learning model accuracy evaluation (precision, recall, F1-score), and AR rendering reliability validation to ensure that susceptibility classification, flat-surface detection, and end-to-end data flow operate accurately and consistently;
5. evaluate the system's overall level of user satisfaction in terms of accessibility, reliability, usability, security, maintainability, and portability of the system using the ISO 25010.

Theoretical Framework

The theoretical framework of FloodSense illustrates how machine-learning-based flood susceptibility classification and markerless augmented reality visualization are integrated into a single mobile platform to deliver location-specific, barangay-level flood susceptibility information to residents of Bacoor City. All user account management and susceptibility lookup data are handled through a centralized Firebase backend, which securely stores user profiles, saved preferences, and pre-computed classification outputs providing on-demand access to flood susceptibility information without requiring a live machine learning inference server, though an internet connection is required to retrieve pre-computed results from Firebase Firestore.

Authentication and Onboarding Module. This module manages secure user registration and login using a username and password credential system. Since Firebase Authentication does not natively support username-based login, the

application implements a standard workaround: upon registration, the user provides a username and password, which are stored in Firebase Firestore, and the system internally generates a surrogate email address used exclusively by Firebase Authentication in the background. The user never interacts with or sees this internal email — the visible login experience is purely username and password. A mandatory onboarding screen presents the application's purpose, scope limitations, and a disclaimer stating that susceptibility outputs are modeled estimates based on environmental features and user-selected rainfall parameters, not real-time flood guarantees. Instructions for flat-surface AR usage are also delivered here before the user proceeds to the main interface.

PAG-ASA-Aligned Rainfall Input Module. Users specify a rainfall scenario through two plain-language dropdown menus aligned with PAGASA's public advisory classification: rainfall intensity (Light, Moderate, Heavy, Intense and Torrential) and duration (1, 3, 6, 12, 24 hours). The application internally maps these selections to standardized millimeter-per-hour ranges, removing the need for residents to interpret technical meteorological values, a design feature directly validated by the BDRRMO key informant.

Machine Learning Flood Susceptibility Classification Module. A supervised classification model (Random Forest or XGBoost) was trained offline on five environmental features: elevation, historical flood records, and rainfall parameters sourced from CPDCO and BDRRMO. All spatial features were researched-identified from institutional datasets, not collected from community surveys. The trained model classifies each barangay into one of four susceptibility levels: Low, Moderate, High, and Very High. Classification outputs are pre-computed for all supported rainfall intensity/duration combinations and stored as static JSON lookup tables in Firebase Firestore; no live model inference is performed at runtime.

Augmented Reality Visualization Module. This module renders the classification output as a color-coded barangay-level susceptibility map anchored to

any detected flat surface using Vuforia Ground Plane tracking. A surface detection indicator (green for valid surface, red for invalid) prevents map rendering until a suitable plane is confirmed. A color legend is visible within the AR view, and a disclaimer is accessible at all times. The map is displayed at a readable scale, and users may pinch-to-zoom to examine individual barangay labels and color-coded susceptibility levels more closely

Results Summary Module and Decision Support Module. A scrollable list of barangays with color-coded susceptibility badges is displayed, populated directly from the pre-computed lookup table retrieved from Firestore based on the user's saved rainfall preferences. This module provides an at-a-glance overview without requiring the AR interface.

All user-related data and susceptibility lookup tables are stored in Firebase Firestore, with account security managed through Firebase Authentication using the surrogate email workaround described in the Authentication and Onboarding Module. The architecture ensures per-user data isolation and stores only the user's registered username, declared barangay, and rainfall preferences; no GPS coordinates or real-time location data are collected, in compliance with the Data Privacy Act of the Philippines (RA 10173).

Overall, the framework demonstrates how processing of spatial environmental features, pre-computed machine learning classification, and augmented reality display are combined into a single community-accessible platform. By converting institutionally held flood datasets into an intuitive, plain-language mobile tool, the system aims to reduce the incidence of uninformed flood-related decisions, strengthen barangay-level flood susceptibility awareness, and provide residents of Bacoor City with a reliable and accessible decision-support application.

The diagram shows how FloodSense works as a system designed to provide barangay-level flood susceptibility information to residents of Bacoor City. It highlights the flow of data from the user's rainfall parameter selections and institutional

environmental datasets through machine learning classification and augmented reality visualization, all connected via a centralized Firebase backend.

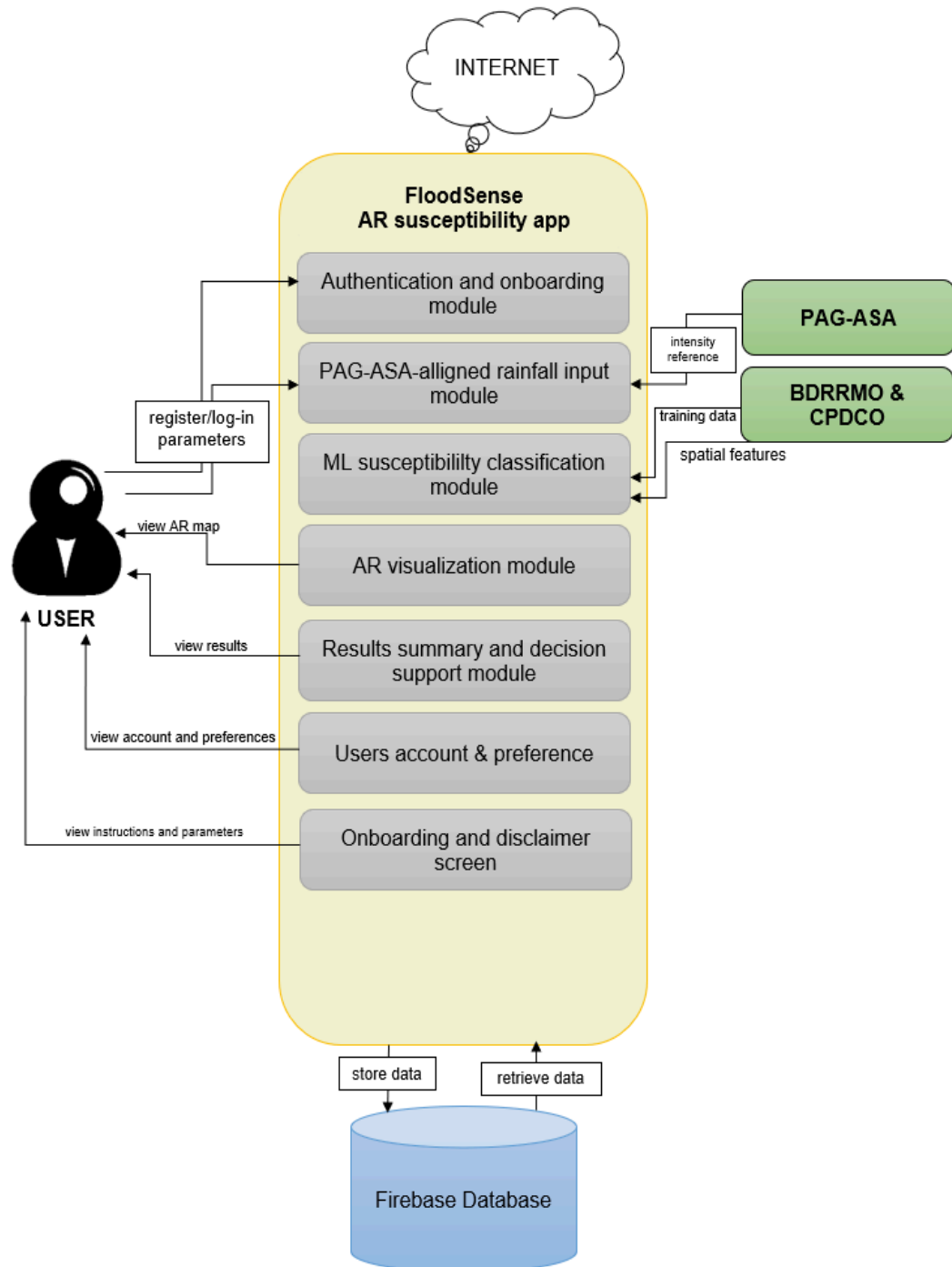


Figure 1. Theoretical Framework of FloodSense: An Augmented Reality Mobile Application with Machine Learning-based Flood Susceptibility Mapping for Bacoor City.

Significance of the Study

The development of FloodSense addresses a critical gap in flood information dissemination by translating institutional hazard data into an intuitive, barangay-level susceptibility map accessible through augmented reality. The following stakeholders stand to benefit directly from the study.

For the **Bacoor Disaster Risk Reduction and Management Office (BDRRMO)**. The application extends the office's capacity to deliver location-specific flood susceptibility information beyond its current channels of social media, email, and barangay coordination. It serves as a complementary decision-support tool that reinforces the office's public awareness and zero-casualty mandates, and opens a pathway for future collaboration by demonstrating how institutional data can be made publicly accessible.

For the City Planning and Development Coordinator Office - Bacoor (CPDCO). The study demonstrates a practical use case for the spatial data the office maintains — including elevation maps, flood hazard maps, and land use classifications — by integrating these datasets into a publicly accessible machine learning model. FloodSense illustrates how institutional planning data, often held only as hardcopy or static outputs, can generate direct community value in the context of disaster risk communication.

For residents of historically flood-affected barangays. The application directly addresses the information deficits uncovered in the survey: 92% of respondents experience flooding, yet only 20% feel well-informed about flood-prone areas, and 64% have made wrong decisions due to insufficient information. By replacing guesswork with a color-coded susceptibility map generated from user-selected rainfall intensity and duration, residents gain an accessible basis for evacuation, route avoidance, and preparedness. The PAGASA-aligned plain-language input removes the technical barrier of estimating rainfall in millimeters,

making the tool usable for community members without specialized training.

For **future researchers**. The study provides a replicable model for integrating machine learning-based susceptibility classification with augmented reality visualization in a Philippine local government context. The documented methodology combining institutional hazard data, open geospatial datasets, and community-validated input design offers a reference framework for similar projects in other flood-prone cities. Additionally, the evaluation of model accuracy, user satisfaction, and adoption feasibility contributes empirical data to the growing body of knowledge on technology-driven disaster risk communication.

Time and Place of the Study

The research and development of the FloodSense application were conducted using the Design Thinking methodology, beginning with data gathering activities prior to the formal study period. The official duration of the study spans from September 2026 to May 2027. All major development activities including system design, machine learning model training, AR integration, and documentation were carried out at the Department of Computer Studies, Cavite State University — Bacoar City Campus, located in Soldiers Hills IV, Molino, Bacoar City, Cavite. The campus served as the primary venue for software development, testing, and academic consultation.

Initial data gathering commenced in early May 2026 during the Empathize phase of Design Thinking. A structured survey was administered to 50 residents across 12 barangays in Bacoar City, many of which have historically experienced flooding, from May 5 to May 12, 2026, to capture prevailing challenges in flood information access, timeliness, and decision-making. On May 12, 2026, a key informant interview was conducted at the Bacoar Disaster Risk Reduction and Management Office (BDRRMO), located at the City of Bacoar Legislative and

Disaster Resilience Building, Bacoar Government Center, Barangay Bayanan, Bacoar City. The interview aimed to validate survey findings, understand the office's current workflows and data holdings, and assess the feasibility of a community-facing flood susceptibility tool. A supplementary key informant interview was also conducted on May 18, 2026 with the Officer-in-Charge of the Bacoar City Planning and Development Coordinator Office (CPDCO), located at the Bacoar Government Center, Barangay Bayanan, Bacoar City. This interview confirmed the availability of spatial datasets including elevation maps, flood hazard maps, and barangay boundary maps and established that these can be released to the research team as hardcopy prints through the existing endorsement letter. The CPDCO interview further validated the environmental and topographic context of Bacoar City as a low-lying catch basin, reinforcing the spatial features selected for the machine learning model. The outputs of the Empathize phase guided the identification of system requirements and informed the overall design direction of the application.

Following data gathering, the study progressed through the Define, Ideate, Prototype, and Test phases within the research timeline. Wireframes, user interface prototypes, and the machine learning model were iteratively refined through consultations with academic advisers and feedback from a pilot pool of target users. From March 2027 to April 2027, the study entered the Test and Iteration phases, during which the developed prototype was evaluated by at least 20 Bacoar City residents and BDRRMO representatives to assess usability, susceptibility classification accuracy, and AR rendering reliability. Feedback collected during this period supported further refinements to improve the application's functionality and user experience.

Scope and Limitation of the Study

The study developed “FloodSense”, a mobile application that provides barangay-level flood susceptibility mapping through augmented reality (AR) visualization and machine learning (ML) classification. The system aimed to equip residents of Bacoar City with an intuitive, data-driven decision-support tool that translates institutional flood hazard data into color-coded susceptibility levels Low, Moderate, High, and Very High rendered over a flat surface via markerless AR ground-plane tracking. Core features included a PAGASA-aligned rainfall input interface using plain-language intensity presets, an onboarding disclaimer, user account management through Firebase (username and password), and a supervised ML model trained on elevation, historical flood records, and user-specified rainfall parameters. The susceptibility classification results were pre-computed for all supported rainfall scenarios and stored as static JSON lookup tables in Firebase Firestore; the mobile application retrieves the appropriate table on demand without a live model server.

The primary users of the application are residents of Bacoar City, particularly those from historically flood-affected barangays. Residents can select a rainfall scenario and view a barangay-level susceptibility map superimposed on any flat surface through their phone camera, allowing them to quickly assess which areas are likely to be affected under their chosen conditions. The map serves as an area-avoidance and preparedness aid, not as a real-time navigation system. The Bacoar Disaster Risk Reduction and Management Office (BDRRMO) is not a direct operational user of the mobile interface but benefits as a partner for data sharing and potential future model retraining collaborations.

The study specified clear operational and geographical boundaries. The system was limited to the territorial jurisdiction of Bacoar City, covering the barangays for which complete historical flood records and spatial datasets could be obtained.

Flood susceptibility classification relied solely on rainfall-induced flooding scenarios; tidal flooding, storm surges, and flash floods originating from upland areas were explicitly excluded from the model's scope and were not visualized by the application. The AR map was anchored to a flat horizontal surface detected by Vuforia's ground-plane engine and did not rely on the user's GPS coordinates — no outdoor positional tracking or real-time location monitoring was performed. Consequently, the application did not function as a real-time flood sensor, a navigation system, or a street-level routing tool. It answered the question *“Which barangays are susceptible at what level under this rainfall scenario?”* rather than providing turn-by-turn directions or live flood extent mapping.

User input was limited to selecting a rainfall intensity from five PAGASA-aligned presets (Light, Moderate, Heavy, Intense, Torrential) and a duration (1, 3, 6, 12, or 24 hours). The application did not accept free-text numerical rainfall values, nor did it automatically pull live weather data from meteorological stations; susceptibility classification depended entirely on the user-selected scenario. Spatial features (elevation) were pre-processed and embedded in the ML model — users were not required to input any location-specific environmental data. The machine learning model was trained on a combination of BDRRMO historical flood records, CPDCO elevation data and land use maps, and its predictions were confined to the four discrete susceptibility classes. Model outputs represented susceptibility likelihood based on static environmental conditions and did not account for dynamic variables such as real-time drainage blockages, soil type, river water levels, or actual rainfall accumulation beyond the user-selected parameters.

The machine learning model intentionally excludes drainage proximity and soil type as input features. The CPDCO interview confirmed that drainage infrastructure in Bacoor City is frequently compromised by garbage accumulation and siltation, making a static drainage map an unreliable predictor of actual flood behaviour. Soil permeability was similarly excluded because a large portion of the

city's surface is paved or built-up, which reduces the influence of natural soil type on flood susceptibility. The model therefore relies on elevation, historical flood records, and user-specified rainfall parameters — the most stable, measurable, and locally validated environmental predictors available.

A methodological limitation is acknowledged in the construction of the training dataset: historical flood records, elevation map, were used both to derive the target susceptibility class labels and as an input for the model. This overlap introduces a risk of data leakage, which may inflate performance metrics during evaluation. The model's generalizability to future or previously unseen rainfall events should be interpreted with this constraint in mind, and subsequent iterations are recommended to employ temporally independent training and validation splits to mitigate leakage.

The study also acknowledged several technical and operational limitations. The application was developed exclusively for the Android operating system and is not yet supported or implemented on iOS devices. The application required an active internet connection to retrieve susceptibility lookup tables from Firebase Firestore and to synchronize user accounts; offline functionality was not implemented and remains a stated limitation. Furthermore, CPDCO stated that it maintains elevation, land use, hazard, and barangay boundary maps, these are available only as hard-copy prints; GIS shapefiles and raster files are classified as confidential and could not be released. Minimum mobile device specifications included Android 10 or higher, 4 GB RAM, 500 MB of free storage, an accelerometer and gyroscope for AR tracking, and a rear camera with autofocus capability. The AR module relied on Vuforia's ground-plane detection, which required a sufficiently textured, flat surface; pointing the camera at uneven ground, or low-texture surfaces could prevent the map from rendering. Onboarding instructions and a green/red surface detection indicator were incorporated to mitigate this limitation, but successful map display ultimately depended on the user positioning the device correctly. Data privacy constraints were observed in accordance with Republic Act 10173, the application collected no GPS

coordinates or personal location data, and user account information was protected through Firebase Security Rules and informed consent during pilot testing.

No admin dashboard was developed, as the study's core contribution focused on integrating machine-learning-based susceptibility classification with markerless augmented reality for community users. A dedicated administrative interface would have required substantial additional development for role-based access, data validation, and upload workflows, diverting effort from the end-user features that directly address the documented information gap. Maintainability is instead provided through a documented offline update pipeline: new flood records can be incorporated by retraining the model and uploading the regenerated lookup tables to Firebase Firestore, a process that requires only basic scripting knowledge and no changes to the mobile application. A user-friendly administrative dashboard for BDRRMO personnel remains a recommended future enhancement, supported by the system's modular architecture. User authentication is limited to username-and-password login via Firebase; social login providers (Google, Facebook) and phone-based OTP verification are not supported. Only one set of rainfall preferences can be saved per user, and multiple saved scenarios are not available.

In terms of data inputs, the application does not integrate real-time rainfall sensors or live weather APIs. All susceptibility classifications are based solely on the user's selected PAGASA-aligned presets rather than current meteorological conditions. The AR visualization is confined to a miniature top-down map anchored to a detected flat surface; GPS-anchored outdoor AR, location-based services, and any form of geographic positioning of the user are not implemented. The AR rendering itself is a flat, color-coded 2D overlay, with 3D terrain models, flood animations, and vertical surface tracking excluded.

The application provides barangay-level susceptibility information only and does not offer navigation arrows, turn-by-turn routing, or real-time flood depth

tracking. The results list displays all barangays without sorting, filtering, or search functionality. Classification output represents physical flood susceptibility only; vulnerability or risk scoring that would combine hazard data with population exposure or community capacity is not within the study's scope.

Finally, the evaluation was conducted through usability testing with a small pilot group of respondents drawn from the original survey pool, and the generalizability of the findings beyond Bacoar City may require further validation with larger and more diverse user populations.

Definition of Terms

The following terms are defined operationally as they are used in this study to ensure clarity and consistency throughout the research:

Augmented Reality (AR). A technology that superimposes digital information—such as images, 3D models, or text—onto a real-world view through a device's camera. In FloodSense, AR is used to render a color-coded flood susceptibility map as a ground-plane overlay on any detected flat surface.

Color-Coded Susceptibility Levels. A visual representation scheme used in the AR map where green indicates Low, yellow indicates Moderate, orange indicates High, and red indicates Very High flood susceptibility.

CPDCO (City Planning and Development Coordinator Office). The local government unit that maintains land use maps and spatial development data for Bacoar City.

Firebase. A Google cloud platform providing backend services including Authentication and Firestore database. FloodSense uses Firebase for user account management, storing preferences, and hosting static content.

Flood Susceptibility. The likelihood or potential of an area to experience flooding based on environmental conditions such as elevation and rainfall.

FloodSense classifies flood susceptibility into four levels: Low, Moderate, High, and Very High.

Ground-Plane Tracking. A markerless AR technique that detects a horizontal flat surface (e.g., floor, table) in the device's camera view and anchors digital content to that surface. FloodSense uses Vuforia's Ground Plane to place the susceptibility map overlay.

Historical Flood Records. Past data documenting flood events, affected areas, flood depth, and frequency. BDRRMO supplied historical flood records for Bacoor City, which served as ground truth for training the machine learning model.

Machine Learning (ML). A branch of artificial intelligence that enables systems to learn patterns from data and make predictions without explicit programming. FloodSense employs a supervised classification model (Random Forest or XGBoost) trained on environmental features to predict barangay-level flood susceptibility.

Markerless AR. An augmented reality method that does not require a physical image target or QR code; it uses device sensors and visual analysis to detect surfaces. Vuforia's ground-plane tracking is markerless, allowing users to place the map on any flat surface.

Onboarding and Disclaimer Screen. A first-launch interface that explains the app's purpose, limitations, and proper use. It includes a disclaimer stating that susceptibility outputs are not real-time guarantees and that users should always follow official warnings.

PAGASA-Aligned Rainfall Input. A user interface element that allows selection of rainfall intensity and duration using plain-language presets (Light, Moderate, Heavy, Intense and Torrential) matched to PAGASA's public rainfall advisory classification. The app internally converts these selections to mm/hr values for the ML model.

Plain-Language Intensity Presets. Descriptive labels for rainfall intensity (Light, Moderate, Heavy, Intense and Torrential) that replace numerical millimeter readings, making the input accessible to non-technical users. These presets are aligned with PAGASA's rainfall advisory categories.

Python (scikit-learn / XGBoost). The programming language and libraries used to develop and evaluate the machine learning classification model. Scikit-learn is used for model building, while XGBoost is the gradient boosting algorithm considered for susceptibility classification.

Rainfall Parameters. The user-selected combination of rainfall intensity (mm/hr) and duration (hours) that represents a hypothetical rainfall scenario. These parameters are fed into the ML model to generate a corresponding susceptibility map.

Random Forest. A supervised machine learning algorithm that constructs multiple decision trees and aggregates their outputs for classification. It is one of the candidate algorithms for the FloodSense susceptibility model.

Surface Detection Indicator. A visual feedback element that displays green when a suitable flat surface is detected by Vuforia and red when the surface is invalid, preventing map rendering until conditions are met.

Susceptibility Classification. The process of assigning a susceptibility level (Low, Moderate, High, Very High) to each barangay based on the output of the machine learning model. This classification directly drives the AR map color coding.

Unity (uGUI). The game engine and development environment used to build the FloodSense mobile application, including all user interface screens (login, preferences, AR view, results summary) via its uGUI system.

User Account and Preference System. A Firebase-based module that allows users to register, log in, and save rainfall intensity/duration preferences for quicker future access.

Vuforia SDK. A software development kit that provides markerless ground-plane tracking and AR content anchoring. FloodSense uses Vuforia to detect flat surfaces and render the susceptibility map overlay.

XGBoost. An optimized gradient boosting machine learning library known for high performance. It is evaluated alongside Random Forest as the classification algorithm for flood susceptibility.

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